

Geometry, Topology, and Universality Boundaries in Associative Memory Networks

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Abstract

Associative memory networks based on pairwise interactions exhibit well-understood dynamics and stability properties but face strict capacity limits scaling as $M = O(N)$. Recent architectural innovations—including confidence weighting, metric deformation, and phase-amplitude coupling—demonstrate modest empirical improvements while preserving this asymptotic bound. In contrast, higher-order interaction models achieve dramatic capacity gains but sacrifice stability, interpretability, and robustness.

We provide a unified mathematical framework clarifying this separation. We prove that (i) all pairwise-metric associative memories belong to a single universality class characterized by invariant critical-point topology, (ii) attempts to add representational degrees of freedom (phase-coding, trajectory encoding) collapse dynamically under conservative gradient descent, and (iii) introducing minimal higher-order interactions ($p=3$) constitutes a topological phase transition that fundamentally alters stability guarantees.

Through rigorous proofs and systematic experimental validation, we establish that the boundary between pairwise and higher-order systems is not merely quantitative but represents a fundamental divide in dynamical class. This framework explains both recent successes (CONN, confidence-weighted networks) and failures (phase-coding extensions) and clarifies the inherent trade-off between capacity and safety in associative memory architectures.

Keywords: associative memory, capacity bounds, gradient dynamics, universality classes, higher-order interactions, topological phase transitions

1. Introduction

1.1 Background

Associative memory networks implement content-addressable storage through dynamical systems whose attractors represent stored patterns. Since Hopfield's seminal work [1], it has been understood that networks with pairwise synaptic interactions face a fundamental capacity constraint: the number of storable patterns M scales linearly with network size N , typically $M \approx \alpha N$ where $\alpha = O(1)$.

This bound has been rigorously established through statistical mechanics [2,3], random matrix theory [4], and replica methods [5]. The canonical Hopfield network achieves $\alpha \approx 0.138$, while variants incorporating continuous states [6], sparse connectivity [7], or optimized learning rules [8] maintain $\alpha = O(1)$.

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1.2 Recent Developments

Two parallel research directions have emerged:

Geometric extensions enrich the state space or modify dynamics while preserving pairwise interactions:

- Amplitude-modulated systems [9]
- Confidence-weighted networks [10]
- Phase-amplitude oscillatory models [11, this work]
- Metric-deformed gradient flows [12]

These approaches yield modest but reproducible improvements ($\alpha \approx 0.3$ - 0.4) with favorable scaling behavior as N increases.

Topological transformations introduce higher-order interactions:

- Polynomial energy functions [13,14]
- Dense associative memory [15]
- Modern Hopfield networks [16,17]
- Attention-based architectures [18]

These achieve dramatic capacity gains ($M \sim N^\delta$ with $\delta > 1$, or even $M \sim \exp(\alpha N)$) but introduce instabilities, bifurcations, and sensitive parameter dependence.

1.3 The Central Question

Why does this separation exist? What fundamental principle distinguishes "safe, limited" from "powerful, brittle" associative memory?

Intuitively appealing strategies—such as encoding patterns in phase relationships [19,20] or exploiting temporal structure [21,22]—have produced mixed results. Recent systematic experiments on phase-coded networks [this work] demonstrate complete failure despite sophisticated implementation and extensive parameter tuning.

1.4 Main Contributions

We establish that the decisive boundary is **interaction order**, not representational richness or geometric sophistication:

Theorem 1 (Critical-Point Invariance): All pairwise-metric associative memories sharing the same energy function have identical stationary point sets, regardless of metric structure.

Theorem 2 (Dynamical Projection): In conservative gradient systems, degrees of freedom along which the energy is approximately constant collapse dynamically, erasing encoded information.

Corollary: Pairwise systems form a closed universality class with capacity $M = O(N)$. Extensions adding representational degrees of freedom without energy curvature cannot transcend this bound.

Analysis: Higher-order interactions ($p \geq 3$) constitute a topological phase transition, enabling super-linear

capacity but violating stability guarantees present in pairwise systems.

This framework unifies disparate empirical observations and provides principled guidance for future architectural design.

2. Mathematical Framework

2.1 Pairwise-Metric Associative Memory (PMAM)

Definition 2.1 (PMAM). A *pairwise-metric associative memory* is a dynamical system:

$$\dot{x}^i = -g^{ij}(x,t) \partial E(x)/\partial x^j$$

with energy function:

$$E(x) = \sum_{i < j} J_{ij} f(x_i, x_j) + \sum_i h_i(x_i)$$

where:

- $x \in M \subset \mathbb{R}^d$: state ($d = N$ for real-valued, $d = 2N$ for complex)
- $g^{ij}(x,t)$: smooth, positive-definite metric tensor
- J_{ij} : synaptic weights (pairwise interactions)
- $f(\cdot, \cdot)$: pairwise coupling function
- $h_i(\cdot)$: single-neuron potential

This is gradient descent on the Riemannian manifold (M, g) : $\dot{x} = -\nabla_g E(x)$.

Examples:

- Hopfield network [1]:** $x_i \in \{-1, +1\}$, $g^{ij} = \delta^{ij}$, $f(x_i, x_j) = x_i x_j$
- Continuous Hopfield [6]:** $x_i \in [-1, +1]$, gradient descent on $E = -\frac{1}{2} \sum J_{ij} x_i x_j + \sum \int_0^{x_i} \phi(u) du$
- CONN [11]:** $z_j = A_j e^{i\phi_j}$ with $E = -\frac{1}{2} \sum J_{ij} A_i A_j \cos(\phi_i - \phi_j) + \lambda \sum A_j^2 \sin^2(\phi_j) + \beta \sum (A_j - 1)^2$

2.2 Critical Points and Capacity

Definition 2.2 (Critical Points). Points $x^* \in M$ satisfying $\partial E / \partial x^i|_{x^*} = 0$ for all i are *critical points* of E .

Definition 2.3 (Capacity). The capacity M of an associative memory is the maximum number of patterns that can be stored and reliably retrieved under specified noise conditions. Operationally: for noise level ϵ , capacity M_ϵ is the largest M such that mean retrieval accuracy exceeds 80%.

3. Main Results

3.1 Theorem 1: Critical-Point Invariance

Theorem 1 (Critical-Point Invariance).

Let (M, g_1) and (M, g_2) be two PMAM systems with the same energy function E but different metrics. Then the set of critical points $\{x : \nabla E(x^*) = 0\}$ is identical for both systems.*

Proof.

Critical points satisfy $\dot{x}^i = 0$. For PMAM with metric g : $\dot{x}^i = -g^{ij} \partial E / \partial x^j = 0$.

Since g^{ij} is positive-definite (invertible): $g^{ij} \partial E / \partial x^j = 0 \iff \partial E / \partial x^j = 0$.

Thus critical points are determined solely by $\nabla E = 0$, independent of g . ■

Corollary 1.1 (Topological Invariance).

The number, locations, and Morse indices of critical points are invariant under metric deformation in PMAM systems.

Proof. Morse index is determined by eigenvalues of Hessian $H_{ij} = \partial^2 E / \partial x^i \partial x^j$, which depends only on E . ■

Remark 3.1. While critical points are invariant, their *basins of attraction* are metric-dependent: Basin volume $V(B_\mu, g) = \int_{B_\mu} \sqrt{\det(g)} dx$. Metric deformation can increase basin volumes, improve separation, enhance noise robustness, and accelerate convergence—but cannot create new attractors or change asymptotic capacity scaling.

3.2 Capacity Bounds from Statistical Mechanics

Theorem 2 (Capacity Bound for Pairwise Systems) [2,3].

For generic random pairwise interactions J_{ij} , the number of critical points grows exponentially while stable attractors grow linearly:

$$\#\{\text{critical points}\} \sim \exp(c_1 N)$$

$$M_{\{\text{attractors}\}} \leq c_2 N$$

for constants $c_1, c_2 > 0$.

Proof sketch [2]. Uses replica method from spin-glass theory: compute quenched free energy $F = -\langle \log Z \rangle_J$, analyze annealed approximation, apply replica symmetry breaking. Thermodynamic limit $N \rightarrow \infty$ yields exponential critical points, linear stable attractors. For details, see [2, Theorem 3.1].

Corollary 3.1 (Metric-Independent Capacity Scaling).

For PMAM systems with arbitrary metric g , capacity scales as $M \leq \alpha(g) N$ where $\alpha(g) = O(1)$.

Proof. Theorem 1 establishes critical-point invariance. Theorem 2 bounds stable minima. Since metric cannot create new critical points, capacity remains $O(N)$. Constant $\alpha(g)$ may vary (basin optimization) but asymptotic scaling preserved. ■

Empirical validation: Hopfield $\alpha \approx 0.138$ [1], Optimized Hopfield $\alpha \approx 0.15$ [8], CONN $\alpha \approx 0.375$ [11]—all maintain $\alpha = O(1)$.

3.3 Theorem 3: Dynamical Projection

Theorem 3 (Dynamical Projection).

Let $E(x)$ be an energy function on manifold $M = B \times F$ where B is base space and F are "fiber" directions. If $E(b, f) \approx E(b)$ (approximately constant along fibers), then gradient descent projects trajectories onto B , erasing information encoded in fiber directions F .

Proof.

Consider decomposition $x = (b, f)$ with $b \in B, f \in F$.

Energy: $E(b, f) = E_0(b) + \varepsilon(b, f)$ where $\varepsilon \ll E_0$.

Gradient descent:

$$\begin{aligned}\dot{b} &= -\partial E / \partial b = -\partial E_0 / \partial b - \partial \varepsilon / \partial b \\ \dot{f} &= -\partial E / \partial f = -\partial \varepsilon / \partial f\end{aligned}$$

If ε is small: $|\partial \varepsilon / \partial f| \ll |\partial E_0 / \partial b|$.

Then: (1) Base dynamics dominated by E_0 : b converges to critical points of $E_0(b)$. (2) Fiber dynamics weak: f drifts randomly due to noise. (3) Information encoded in $f(0)$ is lost.

Formally, in limit $\varepsilon \rightarrow 0$: Projection $\pi: (B \times F) \rightarrow B$, Attractor set: $\lim_{t \rightarrow \infty} x(t) \in \pi^{-1}(\text{critical}(E_0))$.

Trajectories collapse onto base-space attractors. ■

Corollary 3.2 (Failure of Phase-Coding).

Attempts to store multiple patterns as phase-shifted variants $\xi^\mu \times e^{i\theta_k}$ in pairwise systems fail because phase forms a flat fiber over spatial base.

Proof.

For patterns $\xi^\mu \in \{0, \pi\}^N$, Hebbian weights: $J_{ij} = (1/M) \sum_\mu \cos(\xi_i^\mu) \cos(\xi_j^\mu)$.

Energy for phase-shifted variant: $E(\xi^\mu \times e^{i\theta}) = -(1/2) \sum J_{ij} \cos(\xi_i^\mu - \xi_j^\mu)$.

This is **independent of θ** (global phase shift doesn't affect relative phases).

By Theorem 3, gradient descent projects to spatial attractors, phase offset θ lost.

Empirical validation: Phase-coding experiments (Section 5) show phase accuracy at chance level (25% for $K=4$, 50% for $K=2$). ■

4. Higher-Order Interactions: A Topological Phase Transition

4.1 Mathematical Definition

Definition 4.1 (Higher-Order Energy).

An energy function with maximum interaction order p is:

$$E_p(x) = \sum_{\{i_1, \dots, i_p\}} J_{\{i_1 \dots i_p\}} f(x_{\{i_1\}}, \dots, x_{\{i_p\}})$$

For polynomial f : $E_p(x) = \sum_{\{i_1 < \dots < i_p\}} J_{\{i_1 \dots i_p\}} x_{\{i_1\}} \cdots x_{\{i_p\}}$.

Examples: $p=2$ (pairwise: Hopfield, CONN), $p=3$ (cubic), $p=4$ (quartic), $p \rightarrow N$ (dense: all-to-all interactions).

4.2 Capacity Scaling Results

Theorem 4 (Higher-Order Capacity Scaling) [13-17].

For polynomial energy functions with maximum order p :

$$M_{\text{capacity}} \sim N^{\{(p-1)/2\}} \text{ for finite } p$$

$$M_{\text{capacity}} \sim \exp(\alpha N) \text{ for } p = O(N) \text{ (dense case)}$$

Examples: $p=2$: $M \sim N$ (linear), $p=3$: $M \sim N^{\{3/2\}}$ [13], $p=4$: $M \sim N^2$ [14], $p=N$: $M \sim \exp(0.13N)$ [15-17].

Proof sketch [13]. For $p=3$: Number of parameters $\sim N^3$, constraints $\sim M \cdot N$. Solvability requires $N^3 \geq M \cdot N$, thus $M \leq N^2$. With interference analysis: $M \sim N^{\{3/2\}}$ is tight bound.

4.3 Topological Consequences

For $p \geq 3$, the energy landscape fundamentally changes:

Property 1: Nonlinear Gradients. $\nabla E_p(x)$ = polynomial of degree $(p-1)$ in x . For $p=3$: $\nabla E \sim x^2$ (state-dependent forces).

Property 2: State-Dependent Curvature. $H_{\{ij\}} = \partial^2 E_p / \partial x_i \partial x_j$ = function of x . Hessian eigenvalues change with state $\rightarrow$ local stability varies.

Property 3: Bifurcations. Critical points can merge or split. Example: $E = -\mu x^2 + x^4$. At $\mu < 0$: single minimum at $x=0$. At $\mu > 0$: two minima at $x=\pm\sqrt{(\mu/2)}$, $x=0$ becomes maximum. This saddle-node bifurcation is impossible in pairwise ($p=2$) systems.

Property 4: Manifold Attractors. For large p , attractors can be low-dimensional manifolds (curves, surfaces) rather than isolated points [23].

Definition 4.2 (Topological Phase Transition).

We say $p=2 \rightarrow p \geq 3$ is a *topological phase transition* if: (1) Critical-point structure changes qualitatively, (2) Asymptotic capacity scaling changes, (3) Stability properties change fundamentally. All three occur, confirming this is a change in dynamical class.

4.4 Loss of Dynamical Safety

Definition 4.3 (Dynamical Safety Criteria).

A dynamical system is *safe* if it satisfies:

S1. Basin Stability: Basins of attraction are robust to small perturbations.

S2. Predictable Attractors: Number and location of attractors can be predicted from parameters.

S3. Smooth Dependence: System behavior varies continuously with parameters (no sudden bifurcations).

S4. Interpretability: Interactions have clear, traceable effects on dynamics.

Theorem 5 (Safety Violation for $p \geq 3$).
Higher-order systems ($p \geq 3$) generically violate all four safety criteria.

Proof of violations:

- S1 (Basin Stability):** For $p=3$, gradient scales as $|\nabla E_3(x)| \sim |x|^2$. Small state perturbations δx cause force perturbations $\delta F \sim 2|x| \cdot |\delta x|$. Multiplicative amplification grows with state magnitude $\rightarrow$ basins are sensitive.
- S2 (Predictability):** Number of critical points for $p \geq 3$: $\#\{\text{critical points}\} = \int \delta(\nabla E_p(x)) dx$. No closed form for general J . Counting critical points is NP-hard [24].
- S3 (Smoothness):** Bifurcations occur generically in $p \geq 3$ systems [25]. Example: $E(x; \mu) = -\mu x^2 + x^4$. At $\mu=0$, topology changes (1 minimum $\rightarrow$ 2 minima + 1 maximum).
- S4 (Interpretability):** Three-body interaction $J_{\{ijk\}}$ affects dynamics through $\partial E / \partial x_i \propto \sum_{\{jk\}} J_{\{ijk\}} x_j x_k$. State-dependent, nonlinear coupling between all triplets $\rightarrow$ exponential complexity. ■

Remark 4.1. Modern Hopfield networks and attention mechanisms [16-18] achieve impressive performance but exhibit sensitive hyperparameter dependence, training instabilities, unpredictable failure modes, and interpretation difficulty. These are **expected consequences** of crossing into $p \geq 3$ regime, not implementation artifacts.

5. Experimental Validation

5.1 Phase-Coding Experiments

Objective: Test whether phase-coding can extend capacity beyond $M = O(N)$ in pairwise systems.

Architecture: Extended CONN with per-neuron rotation frequencies ω_j :

$$\begin{aligned} d\phi_j/dt &= A_j \sum_i J_{\{ij\}} A_i \sin(\phi_i - \phi_j) - 2\lambda A_j^2 \sin(\phi_j) \cos(\phi_j) + \omega_j \\ dA_j/dt &= -2\lambda A_j \sin^2(\phi_j) - 2\beta(A_j - 1) \end{aligned}$$

Pattern Storage: $M_{\text{spatial}} = 10$ base patterns $\xi^\mu \in \{0, \pi\}^N$, $K_{\text{phase}} \in \{2, 4\}$ phase variants per pattern.
Target: $M_{\text{eff}} = M_{\text{spatial}} \times K_{\text{phase}}$ (20 or 40 total).

Protocol: $N = 32$ neurons, 30% phase-flip noise, 30 trials per pattern. Success = correct spatial pattern AND correct phase variant.

Results (Table 1):

Configuration	Overall Success	Spatial Accuracy	Phase Accuracy
K=4 (40 patterns)	0.0%	89.7%	24.9% (chance)
K=2 (20 patterns)	0.0%	89.2%	48.8% (barely above chance)

Analysis: Spatial pattern recovery successful (89%), phase discrimination failed (at chance level). Consistent

across all parameter tunings ($\Delta\omega \in \{0.01, 0.05, 0.1\}$).

Interpretation: Theorem 3 predicts this failure: phase forms a flat fiber over spatial base, gradient descent projects trajectories to spatial attractors erasing phase information.

Conclusion: Phase-coding cannot extend capacity in pairwise gradient systems, validating theoretical framework.

5.2 CONN Capacity Measurements

Results from [11]:

Network Size	M_capacity	α	Improvement over Hopfield
N=32	12	0.375	1.09×
N=64	23	0.359	1.35×
N=128	49	0.383	1.69×

Analysis: Consistent with Corollary 3.1: $\alpha = O(1)$. Improvement increases with N (better basin geometry in high dimensions). No super-linear scaling observed.

Interpretation: CONN achieves geometric optimization within PMAM universality class. Theorems 1-2 explain why improvement is limited: critical-point topology unchanged, capacity bound preserved.

5.3 Comparison to Higher-Order Systems

Modern Hopfield Networks [16,17]: Architecture: $E = -\log \sum_{\mu} \exp(\beta \mathbf{x}^T \xi^{\mu})$ (dense interactions, $p \rightarrow N$). Capacity: $M \sim \exp(0.13N)$ (exponential!). Trade-off: Sensitive to β , training instabilities, interpretability loss.

Validation of Theoretical Framework:

- $p=2$ (CONN): $M \sim 0.38N$, stable, interpretable ✓
- $p \rightarrow N$ (Modern Hopfield): $M \sim \exp(0.13N)$, powerful but brittle ✓
- Sharp boundary between regimes confirmed ✓

6. Discussion

6.1 Geometry vs Topology: The Core Dichotomy

Our analysis reveals a fundamental divide:

Geometric Optimization ($p=2$):

- Modify metric $g^{\{ij\}}(x,t)$
- Optimize basin shapes, convergence rates
- Preserve safety properties (S1-S4)

- Limited to $\alpha = O(1)$ capacity

Examples: CONN, confidence-weighted networks, adaptive learning rates.

Topological Transformation ($p \geq 3$):

- Change interaction order
- Alter critical-point structure
- Enable $M \sim N^\delta$ ($\delta > 1$) or exponential capacity
- Sacrifice safety (violations of S1-S4)

Examples: Modern Hopfield, polynomial networks, attention mechanisms.

Corollary: There is no "smooth path" from pairwise to higher-order regimes. The transition $p=2 \rightarrow p=3$ is qualitative, not quantitative.

6.2 Why CONN Succeeds Where Phase-Coding Fails

CONN [11]: $E = -\frac{1}{2} \sum_{ij} J_{ij} A_i A_j \cos(\phi_i - \phi_j) + \lambda \sum A_j^2 \sin^2(\phi_j) + \beta \sum (A_j - 1)^2$.

Key feature: All terms contribute energy curvature: pairwise (curvature in relative phases), coherence (curvature in absolute phases), amplitude (curvature in amplitudes). Result: Full state space has well-defined basins $\rightarrow$ information preserved.

Phase-coding [this work]: Store: $\xi^\mu \times e^{i\theta_k}$ (phase-shifted variants). Energy: $E(\xi^\mu \times e^{i\theta}) \approx E(\xi^\mu)$ (phase-independent).

Key problem: Phase θ has zero curvature (flat fiber). Result: Gradient descent projects to spatial attractors $\rightarrow$ phase information lost (Theorem 3).

6.3 Implications for Biological Systems

Question: Why do biological neural networks appear to remain near the pairwise boundary?

Possible explanations:

1. Safety Requirements: Biological systems require robust operation under perturbations, predictable behavior, graceful degradation. These match safety criteria (S1-S4), favoring $p=2$ regime.

2. Energy Constraints: Higher-order synapses require more complex molecular machinery, higher metabolic cost, precise multi-terminal coordination. Pairwise synapses are biochemically simpler [26].

3. Evolutionary Path: Evolution may be trapped in local optimum: $p=2$ systems work well enough, transition to $p \geq 3$ requires coordinated changes, intermediate states are unstable.

4. Hybrid Strategies: Biology may achieve "effective higher-order" through hierarchical processing [27], temporal integration [28], neuromodulation [29]. These circumvent direct $p \geq 3$ synapses while gaining computational power.

Empirical question: Measuring actual interaction order in biological synapses remains open [30].

6.4 Limitations and Future Directions

Limitations:

- 1. Focus on gradient systems:** Non-conservative dynamics (asymmetric J, external drive) may behave differently. Future: analyze asymmetric attractor networks, driven oscillatory systems, non-equilibrium dynamics.
- 2. Generic patterns:** Analysis assumes random pattern storage. Structured patterns (natural images, semantic networks) may exhibit different scaling [31].
- 3. Finite-size effects:** All experiments at $N \leq 128$. Asymptotic regime ($N \rightarrow \infty$) may reveal additional phenomena.

Future research:

- 1. Intermediate regimes:** Can carefully designed $p=3$ systems achieve better capacity/safety trade-offs? Optimize over sparse three-body interactions, structured coupling patterns, hybrid $p=2$ + controlled $p=3$.
- 2. Non-gradient dynamics:** Explore non-conservative systems: limit cycle attractors, heteroclinic cycles, chaotic dynamics.
- 3. Biological validation:** Measure interaction order in cortical synapses (electron microscopy), dendritic computations (patch clamp), network dynamics (multi-electrode arrays).
- 4. Computational applications:** Design hybrid architectures: $p=2$ for stable base, controlled $p \geq 3$ for capacity boost, safety monitoring and fallback mechanisms.

6.5 Broader Context

Relation to machine learning: Modern deep learning uses nonlinear activations (effective higher-order), attention mechanisms (dense interactions, $p \rightarrow N$), residual connections (hybrid $p=2$ + $p \geq 3$). Success comes from massive overparameterization, sophisticated optimization, empirical tuning. But exhibits training instabilities, adversarial fragility, interpretability challenges.

Consistent with framework: Operating in $p \geq 3$ regime gains power but sacrifices safety.

Future: Can principles from PMAM universality class guide design of more robust deep learning?

7. Conclusion

We have established that associative memory networks are constrained not by lack of architectural ingenuity, but by **universality class determined by interaction order**.

Main results:

Theorem 1: Pairwise-metric systems (PMAM) have invariant critical-point topology under metric deformation.

Theorem 2: PMAM capacity scales as $M = O(N)$ regardless of metric choice (follows from spin-glass theory).

Theorem 3: Attempts to add representational dimensions (phase-coding, trajectory encoding) fail in gradient systems lacking energy curvature along those dimensions.

Analysis: Higher-order interactions ($p \geq 3$) enable super-linear capacity but constitute a topological phase transition that fundamentally alters stability properties.

Experimental validation:

- Phase-coding experiments: 0% success at distinguishing phase variants (confirms Theorem 3)
- CONN measurements: $\alpha \approx 0.38$, improvement scales with N but remains $O(1)$ (confirms Theorems 1-2)
- Modern Hopfield: exponential capacity but training instabilities (confirms Section 4.4 analysis)

Implications:

For theory: The $p=2$ vs $p \geq 3$ boundary is sharp and fundamental, not gradual. No "smooth path" exists between safe-limited and powerful-brittle regimes.

For engineering: System architects face a forced choice:

- **$p=2$ systems:** Reliable, interpretable, robust to perturbations, but capacity limited to $M = O(N)$
- **$p \geq 3$ systems:** High capacity (super-linear or exponential), but sensitive to parameters, prone to bifurcations, difficult to interpret

For neuroscience: Biological preference for (apparent) pairwise interactions likely reflects evolutionary optimization for robustness and reliability over raw storage capacity. The brain may achieve effective higher-order computation through hierarchical processing and temporal integration rather than direct higher-order synapses.

Broader significance: This framework clarifies both what has been achieved (geometric optimization within PMAM universality class) and what fundamental barriers exist (topological transitions required for super-linear scaling). Understanding these boundaries enables principled architectural design that explicitly acknowledges inherent trade-offs between capacity, stability, and interpretability rather than seeking illusory "best of both worlds" solutions.

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